Logistic Regression

- Consider a two-class classification problem with the dataset $\{(x_j, y_j) : j = 1, \dots, m\}$, where $x_j \in \mathbb{R}^n$ and $y_j \in \{+1, -1\}$

- our prediction function is of the form

$$\phi(x_j, w) = \frac{1}{1 + e^{w^t x_j}},$$

where $\phi(x, w)$ can be interpreted as the probability that a data point with feature vector x_j belongs to class $+1$.

- $1 - \phi(x_j, w) = \dfrac{e^{w^t x_j}}{1 + e^{w^t x_j}}$ is then the probability that a data point with feature vector x_j belongs to class -1.

data point #	feature vect	labels	likelihood
1	x_1	$+1$	$1/1+e^{w^t \cdot x_1}$
2	x_2	-1	$e^{w^t \cdot x_2}/1+e^{w^t \cdot x_2}$
3	x_3	-1	$e^{w^t x_3}/1+e^{w^t \cdot x_3}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
m	x_m	$+1$	$1/1+e^{w^t \cdot x_m}$

- likelihood of the data set $L(w) = \dfrac{1}{1+e^{w^t \cdot x_1}} \dfrac{e^{w^t x_2}}{1+e^{w^t x_2}} \dfrac{e^{w^t x_3}}{1+e^{w^t x_3}} \cdots \dfrac{1}{1+e^{w^t \cdot x_m}}$

- $\ln L(w) = -\ln(1+e^{w^t x_1}) + w^t x_2 - \ln(1+e^{w^t x_2}) + w^t x_3 - \ln(1+e^{w^t x_3}) \cdots - \ln(1+e^{w^t \cdot}$

$$= -\sum_{j=1}^{m} \ln(1+e^{w^t \cdot x_j}) + \sum_{j=1}^{m} \mathbb{1}(y_j = -1) \cdot w^t \cdot x_j$$

- $$\max_{w \in \mathbb{R}^n} \left\{ -\sum_{j=1}^{m} \ln(1+e^{w^t x_j}) + \sum_{j=1}^{m} \mathbb{1}(y_j = -1)\, w^t \cdot x_j \right\}$$

Multi-Class Logistic Regression

- Consider a multi-class classification problem where the dataset is of the form $\{(x_j, y_j) : j=1,\dots,m\}$ where $x_j \in \mathbb{R}^n$ and $y_j \in \{1,\dots,K\}$

- Our prediction function has the form:

$$\phi_k(x_j, w_1, \dots, w_K) = \frac{e^{w_k^t x_j}}{\sum_{\ell=1}^{K} e^{w_\ell^t x_j}},$$

where $\phi_k(x_j, w_1, \dots, w_K)$ is the probability that a datapoint with feature vector x_j belongs to class k.

- Let's use maximum likelihood estimation to estimate the parameters $(w_1, w_2, \dots, w_K) \in \mathbb{R}^{n \times K}$

Data point	Feature vect.	Label	likelihood
1	x_1	3	$e^{w_3^t \cdot x_1} \Big/ \sum_{l=1}^{K} e^{w_l^t \cdot x_1}$
2	x_2	1	$e^{w_1^t \cdot x_2} \Big/ \sum_{l=1}^{K} e^{w_l^t \cdot x_2}$
3	x_3	4	$e^{w_4^t \cdot x_3} \Big/ \sum_{l=1}^{K} e^{w_l^t \cdot x_3}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
M	x_M	3	$e^{w_3^t x_M} \Big/ \sum_{l=1}^{K} e^{w_l^t \cdot x_M}$

$$L(w_1, w_2, \dots, w_K) := \frac{e^{w_3^t x_1}}{\sum_{l=1}^{K} e^{w_l^t x_1}} \cdot \frac{e^{w_1^t x_2}}{\sum_{l=1}^{K} e^{w_l^t x_2}} \cdot \frac{e^{w_4^t x_3}}{\sum_{l=1}^{K} e^{w_l^t x_3}} \cdots \frac{e^{w_3^t \cdot x_M}}{\sum_{l=1}^{K} e^{w_l^t x_M}}$$

$$\ln L(w_1, w_2, \dots, w_K) = w_3^t x_1 - \ln\left(\sum_{l=1}^{K} e^{w_l^t x_1}\right) + w_1^t x_2 - \ln\left(\sum_{l=1}^{K} e^{w_l^t \cdot x_2}\right)$$

$$+ w_4^t x_3 - \ln\left(\sum_{l=1}^{K} e^{w_l^t x_3}\right) \cdots + w_3^t x_M - \ln\left(\sum_{l=1}^{K} e^{w_l^t x_M}\right)$$

$$= \sum_{j=1}^{M} \sum_{k=1}^{K} \mathbb{1}(y_j = k)\, w_k^t x_j - \sum_{j=1}^{M} \ln\left(\sum_{l=1}^{K} e^{w_l^t \cdot x_j}\right)$$

$$\max_{(w_1,\dots,w_K)\in\mathbb{R}^{n\times K}} \left\{ \sum_{j=1}^{M} \sum_{k=1}^{K} \mathbb{1}(y_j = k)\, w_k^t x_j - \sum_{j=1}^{M} \ln\left(\sum_{\ell=1}^{K} e^{w_\ell^t x_j} \right) \right\}$$

Features for each data point (Income, # of years in school)

data point	feature vect	label	likelihood
1	(50K, 11)	A	$e^{0.1 \cdot 50 + 0.5 \cdot 11} / (e^{0.1 \cdot 50 + 0.5 \cdot 11} + e^{0.4 \cdot 50 + 0.1 \cdot 11})$
2	(150K, 16)	B	$e^{0.4 \cdot 150 + 0.1 \cdot 16} / (e^{0.1 \cdot 150 + 0.5 \cdot 16} + e^{0.4 \cdot 150 + 0.1 \cdot 16}$

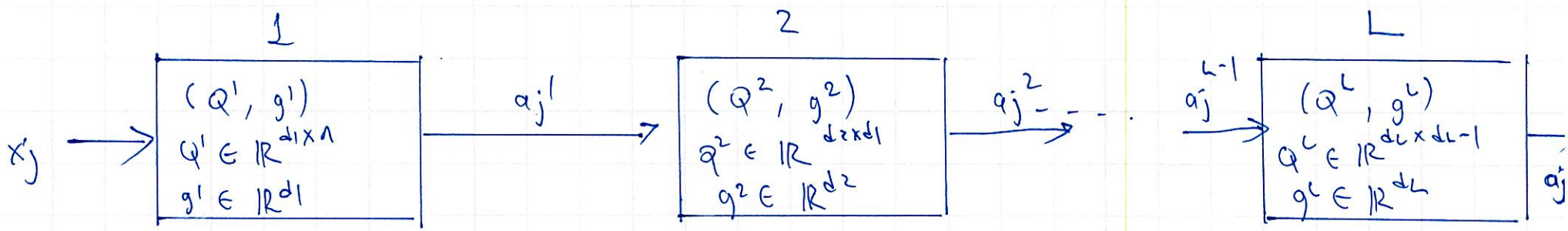
Parameter vectors $(w_A, w_B) = ((w_{A,\text{Income}}, w_{A,\#\text{years}}), (w_{B,\text{Income}}, w_{B,\#\text{of years}}))$

$= ((0.1, 0.5), (0.4, 0.1))$

Deep learning

Consider a multi-class classification problem with datapoints

$\{(x_j, y_j) : j = 1, \dots, m\}$ $\quad x_j \in \mathbb{R}^n \quad y_j \in \{1, 2, \dots, K\}$



$$a_j^1 = \sigma_1(\underbrace{Q^1}_{d_1 \times n} \underbrace{x_j}_{n \times 1} + \underbrace{g^1}_{d_1 \times 1})$$

$(d_1 \times 1)$

$$a_j^2 = \sigma_2(\underbrace{Q^2}_{d_2 \times d_1} \underbrace{a_j^1}_{d_1 \times 1} + \underbrace{g^2}_{d_2 \times 1})$$

$(d_2 \times 1)$

Each component of σ_1 can be sigmoid $\sigma(t) = 1/(1+e^t)$

rectified linear unit $\sigma(t) = \max\{t, 0\}$

Given the parameters $(Q^1, q^1, Q^2, q^2, \dots, Q^L, q^L)$ of the neural network, corresponding to the input x_j in the first layer, the output we get from the last layer is given by

$$\underbrace{\sigma_L\Big(Q^L \underbrace{\sigma_{L-1}\Big(\dots \underbrace{\sigma_2\Big(Q^2 \underbrace{\sigma_1(Q^1 x_j + q^1)}_{\text{output from layer 1}} + q^2\Big)}_{\text{output from second layer}} \dots\dots\Big)}_{\vdots} + q^L\Big)}_{\text{output from last layer}} = Z(x_j, Q^1, q^1, Q^2, q^2, \dots, Q^L, q^L)$$

Solve a multi-class logistic regression problem by using $Z(x_j, Q^1, q^1, Q^2, q^2, \dots, Q^L, q^L)$ as the feature for datapoint j and y_j as the label for data point j.